

Topic 1: Intelligent Controllers in Smart Cities and Sustainable Energy Systems

Introduction

As cities grow and energy demands rise, traditional ways of managing electricity are no longer enough. Smart cities need intelligent systems that can automatically monitor, control, and optimise how energy is produced, distributed, and consumed. The research paper explores how the Internet of Energy (IoE) — a concept that merges the Internet of Things (IoT) with energy systems — enables these intelligent controllers to work effectively across a city's entire energy infrastructure.

What is the Internet of Energy (IoE)?

The IoE draws inspiration from the IoT, where devices, sensors, and machines communicate with each other in real time. In the energy context, IoE integrates various energy sources, storage systems, and distribution networks into one intelligent, self-managing ecosystem. The paper describes IoE as a system designed to "provide critical information for the control and optimisation of the power grid, with the ultimate goal of enhancing its self-sufficiency."

Think of it like a smart thermostat in your home — but scaled to the entire city. Instead of just adjusting the temperature in one room, the IoE adjusts energy flows across thousands of buildings, streetlights, factories, and hospitals, all at once.

Smart Grids: The Backbone of Intelligent Energy Control

The most central intelligent controller discussed in the paper is the **smart grid**. Unlike conventional electricity grids that simply push power from a central plant to consumers, smart grids enable two-way communication between the energy provider and the consumer. They use sensors, meters, and data analytics to monitor what is happening in real time and make automatic adjustments.

Key features of smart grids as intelligent controllers include:

- **Demand-Side Management (DSM):** The paper explains that DSM is a critical part of scheduling optimisation. It involves controlling when and how residential or commercial consumers use electricity. For example, during peak hours when everyone switches on appliances, the system can automatically shift some loads — like running dishwashers or charging electric vehicles — to off-peak hours. This is called **load shifting**, and it prevents overloading the grid.
 - **Smart Meters:** These devices give consumers real-time feedback on their energy usage and carbon emissions, helping them make informed choices. Cities like those in India's Smart Cities Mission (SCM), covering 100 cities, have adopted smart meters as part of their core energy infrastructure.
 - **Peak Demand Management:** Smart grids can identify energy-hungry appliances — air conditioners, water heaters, pool pumps — and control them intelligently to flatten the demand curve.
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Microgrids and Nanogrids: Localised Intelligent Controllers

The paper introduces two important sub-systems of the smart grid: **microgrids** and **nanogrids**. These are smaller, self-contained energy systems that can function independently or in connection with the main grid.

- A **microgrid** typically powers a small group of consumers such as a hospital, university campus, or residential complex. It can use renewable sources like solar panels, wind turbines, micro-turbines, and fuel cells, and has a power capacity of around 10 to 100 kW.
- A **nanogrid** is even smaller, designed for individual residential or commercial buildings, with a power output of up to 25 kW.

Example: Imagine a hospital during a citywide power cut. If it operates on a microgrid with its own solar panels and battery storage, the intelligent controller within the microgrid automatically switches to local energy production, ensuring critical medical equipment stays powered without interruption. This is exactly the kind of resilient, localised control the paper promotes.

Classification of Energy Management Approaches

Section 5 of the paper presents a clear classification of how intelligent controllers manage energy in IoT-driven smart cities. Four main approaches are identified:

1. Scheduling Optimisation This involves planning when energy-consuming tasks or devices should operate. The objective is to reduce overall electricity usage by efficiently managing available resources. In a smart city, this could mean scheduling industrial machinery to run at night when electricity is cheaper and demand is lower.

2. Low-Power Device Transceivers IoT devices in smart cities often have limited power. Intelligent controllers must manage the radio duty cycles of these devices — essentially, how often a device "wakes up" to send or receive data. Researchers focus on minimising these cycles to extend device life and reduce overall energy consumption.

3. Cognitive Framework Since IoT devices vary greatly in size, function, and behaviour, predicting their energy needs is complex. A cognitive framework introduces reasoning and learning capabilities into the system. The paper describes a cognitive management architecture that makes decisions by considering the context of each IoT device — similar to how a human manager would assess a situation before making a call, but done automatically and at high speed.

4. Cloud Computing Technology Cloud platforms enable smart cities to process and store enormous volumes of energy data efficiently. A cloud-based energy management solution allows the system to analyse patterns across thousands of devices and automatically optimise resource use. The paper cites a cloud-based solution introduced for smart buildings to tackle energy management in industrial contexts.

AI and Machine Learning as Intelligent Controllers

Section 9.2 of the paper highlights how Artificial Intelligence (AI) and Machine Learning (ML) take intelligent energy control to the next level. AI algorithms can:

- **Predict maintenance needs** before equipment fails, reducing costly downtime.
- **Forecast future energy demand** based on historical usage data.
- **Manage renewable energy sources** by predicting when solar or wind power will be available and adjusting storage and distribution accordingly.
- **Drive automated demand response** by anticipating demand peaks and automatically shifting loads.

The paper describes AI/ML as collecting and analysing real-time data from smart meters, sensors, renewable sources, and electric vehicles, and using advanced algorithms to "discern intricate patterns, anomalies, and correlations that enable informed decision-making." This is the very definition of an intelligent controller operating at city scale.

The Quantum IoT Architecture: A New Conceptual Framework

One of the paper's original contributions is a proposed **Quantum IoT architecture** for sustainable smart cities (Section 8). This four-layer framework represents the future of intelligent energy control:

- **Physical Layer:** Sensors and actuators that collect real-time data (temperature, humidity, motion, light).
- **Quantum Network Layer:** Manages data transmission using IoT gateways and communication technologies like Wi-Fi, LoRa, Zigbee, Bluetooth, and Satellite.
- **Quantum Teleportation Layer:** Uses quantum repeaters to ensure secure and efficient data transfer over long distances.
- **Application Layer:** Connects users and services to the network, enabling intelligent commands and responses.

This architecture shows where intelligent control is heading — not just smarter grids, but quantum-enabled, ultra-secure, real-time energy systems embedded in the city's very infrastructure.

Conclusion

Intelligent controllers in smart cities, as described by Mishra and Singh (2023), range from smart grids and microgrids to AI-driven demand management and quantum IoT architectures. At every level, the goal is the same: use data, connectivity, and intelligence to reduce waste, increase efficiency, and create a sustainable urban energy future. The IoE provides the framework that ties all of these controllers together into one coherent, responsive system.

Topic 2: AI in Electric Vehicle (EV) Charging Infrastructure

Introduction

Electric vehicles (EVs) are becoming a cornerstone of sustainable urban transportation. As cities aim to reduce carbon emissions and move away from fossil fuels, EVs offer a clean alternative — but only if the energy used to charge them is managed intelligently. Mishra and Singh (2023) discuss EVs not in isolation, but as active participants within the larger Internet of Energy (IoE) ecosystem. From vehicle-to-grid integration to AI-driven traffic management, the paper reveals how artificial intelligence is shaping the way EVs interact with smart city energy infrastructure.

Why EVs Matter in the IoE Framework

The paper positions EVs as important components in addressing climate change, noting that they "contribute to the reduction in carbon emissions." But more than just cleaner transport, EVs in a smart city serve a dual role: they are both consumers of energy and potential sources of stored energy.

The paper describes how "smart grids have the capability to identify and incorporate energy generated or stored by consumers" — including EV owners. This sets the stage for a fundamentally different relationship between vehicles and the power grid, one where AI plays a critical coordination role.

Vehicle-to-Grid (V2G): EVs as Mobile Energy Storage

One of the most significant AI applications related to EV charging discussed in the paper is **Vehicle-to-Grid (V2G) technology**. In a V2G setup, EVs do not just draw power from the grid — they can also push power back into it.

The paper describes V2G as a system where "EVs can serve as mobile energy storage units, capable of contributing energy back to the grid during peak demand or drawing energy during off-peak hours." This **bidirectional energy flow** has major implications:

- During the day when solar energy is abundant, an EV's battery charges up.
- During evening peak hours when demand is high, the EV can supply stored energy back to the grid.
- The AI system manages this exchange automatically, deciding when to charge and when to discharge based on grid conditions and electricity prices.

Example: A fleet of delivery vehicles parked overnight at a depot could all be managed by an AI controller. The system monitors electricity tariffs, grid demand, and each vehicle's schedule. It charges the vehicles cheaply at 2 AM, and if the grid faces a spike in demand at 7 PM, it draws a small amount of energy from each vehicle's battery to assist — all without any human intervention.

The paper cites research (reference [41]) presenting "two power management solutions for enabling V2G technology in grid-associated microgrids with a supervisory power imbalance," confirming that this is an active area of real-world research.

AI and the Smart Grid for EV Integration

The paper's treatment of AI (Section 9.2) is directly relevant to EV charging management. AI algorithms applied to the smart grid can:

- **Optimise EV charging schedules** based on electricity prices, renewable energy availability, and grid stability.
- **Balance the load** created by many EVs charging simultaneously — a significant challenge as EV adoption grows.
- **Integrate EVs into renewable energy management**, ensuring that EVs preferentially charge when solar or wind generation is high, thus maximising the use of clean energy.

The paper states that AI/ML "plays a pivotal role in orchestrating optimised energy distribution, seamlessly integrating factors such as real-time demand, the availability of renewable sources, and grid stability." Applied to EV charging, this means an AI system can decide — in real time — which charging points should activate, at what rate, and for how long.

6G-IoT and Intelligent EV Transportation

Section 9.1 of the paper introduces the role of **6G networks and the IoT** in enabling next-generation EV applications. The paper identifies several energy-efficient applications for the Internet of Vehicles (IoV):

Intelligent Traffic Management By using sensors and data analytics to identify traffic patterns in real time, cities can allow for more efficient traffic flow, which directly reduces fuel and energy consumption. The paper specifically mentions "6G-connected traffic lights, which adjust their timings based on the number of vehicles on the road." For EVs, smoother traffic means less energy wasted on stop-and-go driving.

Predictive Maintenance for Vehicles IoT sensors in EVs can monitor the health of the battery, motor, and other components, predicting maintenance needs before a breakdown occurs. This improves vehicle efficiency and extends the life of the battery — which is the most energy-intensive component of an EV.

Energy-Efficient Route Planning The paper notes that using "real-time traffic data, weather information, and other factors, vehicles can be routed on the most energy-efficient routes, reducing fuel consumption and emissions." For EVs, this also means routing vehicles to charging stations at the right time — neither too early (wasting a trip) nor too late (running out of charge).

Autonomous Driving The paper briefly discusses how 6G-IoT could enable safer and more efficient autonomous driving through vehicle-to-vehicle and vehicle-to-infrastructure communication. Autonomous EVs could be programmed to charge themselves during off-peak hours, optimise routes, and even participate in V2G without any human input.

EV Charging and Demand-Side Management

The paper's discussion of **Demand-Side Management (DSM)** in Section 5.1 is directly applicable to EV charging. DSM involves controlling when electricity is consumed to lower costs and reduce strain on the grid. EV charging is one of the largest flexible loads available in a smart city — meaning it is one of the easiest loads to shift without inconveniencing users significantly.

An AI-powered DSM system for EV charging would:

- Detect when grid demand is low (e.g., late at night) and automatically begin charging EVs.
- Pause or slow charging when demand spikes.
- Use price signals to incentivise EV owners to charge at cheaper, off-peak times.

This type of intelligent scheduling is precisely what the paper describes under its "scheduling optimisation" category of energy management.

Smart Cities Initiative and EV Infrastructure

The paper references India's **Smart Cities Mission (SCM)** across 100 cities, which has a primary goal of equipping urban centres with "high-speed internet connectivity, intelligent transportation systems, and sustainable renewable energy solutions." EVs and their charging infrastructure are a natural part of this vision — connected to smart grids, managed by AI, and powered by renewable energy.

The paper also cites research evaluating "common frameworks for the interaction between intelligent transportation and EVs in smart cities" and exploring "optimal pricing strategies for EV charging stations," showing that the economic and technical dimensions of EV infrastructure are both being actively researched.

Conclusion

While the paper does not dedicate a standalone chapter to EV charging infrastructure, it weaves EV-related AI applications throughout its broader discussion of IoE, smart grids, V2G, 6G-IoT, and demand-side management. The picture that emerges is one where EVs are deeply integrated into the city's intelligent energy system — not just passive consumers of electricity, but active contributors to grid stability, renewable energy utilisation, and sustainable urban living. AI is the intelligence layer that makes all of this coordination possible.

Introduction

Renewable energy sources — solar, wind, hydroelectric, geothermal, and biomass — are at the heart of the global transition to sustainable energy. But unlike conventional fossil fuel plants, renewable energy is intermittent and unpredictable. The sun does not always shine, and the wind does not always blow. This makes intelligent control absolutely essential — without it, integrating renewables into city-wide energy systems would be chaotic and unreliable.

The Evolution of Energy Systems: Where Intelligent Control Fits

The paper maps the history of energy into five levels (Section 3), providing useful context for understanding where intelligent control sits today:

- **Level 1 — Biomass:** Wood and animal waste for cooking. No control systems.
- **Level 2 — Fossil Fuels:** Coal, oil, gas for industry. Centralised but not intelligent.
- **Level 3 — Renewable Energy:** Wind, solar, hydro integrated into the grid. Requires smarter management due to variability.
- **Level 4 — Advanced Digital Technologies (Current):** Smart grids, energy storage, and AI. Intelligent control becomes essential.
- **Level 5 — Future Vision:** Fully decentralised, integrated, intelligent systems with local renewable generation, intelligent infrastructure, and real-time data optimisation.

Today's research, as reviewed in the paper, is primarily at Level 4, with efforts building toward Level 5. Intelligent control for renewable energy is what makes the jump from Level 3 to Level 4 possible.

Recent Research in Smart Grid Control for Renewables

AI and Machine Learning for Renewable Energy Management Section 9.2 of the paper reviews how AI technologies are being used to manage the inherent unpredictability of renewable energy:

- AI predicts **fluctuations in wind and solar energy production** and adjusts storage and distribution accordingly. This is critical because a cloud blocking sunlight can instantly reduce a solar farm's output by 50% — an AI system can anticipate this and draw from battery storage to compensate.
- **Forecasting energy demand** using historical data allows energy companies to plan how much renewable energy to generate or procure in advance.
- AI analyses **market data and weather forecasts** to predict energy prices and optimise energy trading — ensuring that excess renewable energy is sold at the best time and purchased when prices are lowest.

Reinforcement Learning for Island Energy Systems The paper cites a recent study (reference [52]) on a "Hybrid Policy-based Reinforcement Learning (HPRL) adaptive energy management system tailored to achieve optimal performance in an island group energy system facing constraints in energy transmission." This represents cutting-edge research where an AI agent learns — through trial and error — the best policy for managing energy flows in a renewable-powered microgrid, even when it cannot freely transmit energy between all parts of the network.

Graph Convolutional Networks (GCNs) in Energy The paper references research (reference [51]) on the emerging role of Graph Convolutional Networks (GCNs) in the energy sector. GCNs are a type of neural network that can model complex interconnected systems — like a power grid — and optimise energy flows across them. The paper notes their "potential to transform computational intelligence, leading to enhanced energy management and efficiency."

Intelligent Control in Microgrids: Key Case Studies

Case Study 1 — Standalone PV-Wind Hybrid Microgrid The paper directly cites a study (reference [48]) on a standalone microgrid designed for remote areas. The system integrates a **photovoltaic (PV) setup** and a **Wind Energy Conversion System (WECS)** driven by a Permanent Magnet Synchronous Generator (PMSG). The study demonstrates the design and control strategies needed to make such a hybrid system reliable and sustainable for regions far from the main grid.

This is a strong example of intelligent control for renewables: the system must constantly balance the intermittent outputs of both solar and wind sources, manage battery storage, and ensure an uninterrupted power supply — all automatically.

Case Study 2 — Hybrid Storage in DC Nano/Microgrids Another case study cited (reference [47]) focuses on extending DC Bus Signalling and Droop Control for **hybrid storage units** that combine supercapacitors and batteries. Supercapacitors handle rapid, short-term power bursts while batteries manage longer-term storage. The intelligent control system (droop control) coordinates between the two, improving energy storage efficiency and system durability in decentralised DC nano/microgrids. This directly addresses the challenge of making small-scale renewable energy systems more stable and efficient.

Case Study 3 — Event-Triggered Distributed Hybrid Control for Integrated Energy Systems The paper references research (reference [49]) presenting a "novel approach employing event-triggered distributed hybrid control to ensure an Integrated Energy System (IES) operates both securely and economically." Instead of continuously running a controller (which wastes computation and energy), this system only activates when a significant event occurs — a demand spike, a sudden drop in solar output, or a fault. This is a highly efficient approach to intelligent control and represents the direction of current research.

Real-World City Projects: Intelligent Renewable Energy in Practice

The paper dedicates significant space (Table 1, Section 2) to documenting real-world initiatives where intelligent control of renewable energy has been deployed across European and Asian cities:

STEEP (2013–2015) — San Sebastian, Bristol, Florence Explored the integration of technologies like **photovoltaic panels, cogeneration units, geothermal heat pumps, thermal energy storage**, waste heat recovery, and smart meters. The project demonstrated how multiple renewable technologies, when intelligently coordinated, can dramatically reduce a city's fossil fuel dependence.

STEP UP (2012–2015) — Ghent, Glasgow, Gothenburg, Riga Focused on **Combined Heat and Power (CHP)** systems alongside renewables like biomass, geothermal, and solar thermal energy. Intelligent control ensured that heat and electricity were generated in tandem, reducing waste.

ZenN (2013–2017) — Oslo, Malmo, Grenoble Targeted **nearly zero-energy neighbourhoods** by utilising PV panels, solar thermal collectors, heat pumps, and cogeneration units. The intelligent control challenge here was to coordinate all these sources to achieve net-zero energy consumption at the neighbourhood level.

R2CITIES (2013–2018) — Valladolid, Genoa, Istanbul Employed PV, solar thermal, and energy storage systems to move entire cities toward near-zero energy status, demonstrating large-scale deployment of intelligent renewable energy control.

India's Smart Cities Mission (SCM) — 100 Cities, 2015–Ongoing Perhaps the largest ongoing effort cited in the paper. The mission's goal is to equip 100 Indian cities with intelligent transportation, high-speed internet, and **sustainable renewable energy solutions** — integrating IoT-based intelligent control across all urban energy systems.

Energy Harvesting: An Emerging Frontier of Intelligent Control

Section 6 of the paper reviews the latest research in **IoT-enabled energy harvesting** — capturing ambient energy (solar, kinetic, piezoelectric, RF signals) and converting it into usable power for IoT devices. This is a frontier area of intelligent control for renewables:

- **Solar energy harvesting** powers streetlights, cameras, and environmental monitors — all managed by intelligent controllers that switch to battery backup when sunlight is insufficient.
- **Kinetic energy harvesting** from foot traffic or vehicle movement generates electricity in public spaces — an innovative form of distributed renewable generation.
- **RF energy harvesting** captures radio frequency signals and converts them into DC power for IoT sensors, reducing the need for battery replacements across millions of devices.

The paper notes the emergence of the **Internet of Energy Harvesting Things (IoEHT)** as a research concept addressing how to coordinate and intelligently manage these diverse micro-energy sources across a city.

Blockchain for Renewable Energy Transactions

Section 9.6 of the paper covers **blockchain technology** as an emerging tool in intelligent renewable energy control. Blockchain is being used for:

- Managing **renewable energy certificates** to verify that electricity sold to consumers genuinely came from clean sources.
- Enabling **peer-to-peer (P2P) electricity trading**, where a household with rooftop solar panels can sell excess energy directly to a neighbour — automatically, securely, and without a centralised energy company in the middle.
- **Managing EV charging transactions** in a transparent and tamper-proof way.

This is a particularly exciting development because it shifts intelligent energy control from centralised authorities to decentralised, community-level systems — which is the essence of Level 5 in the paper's energy evolution model.

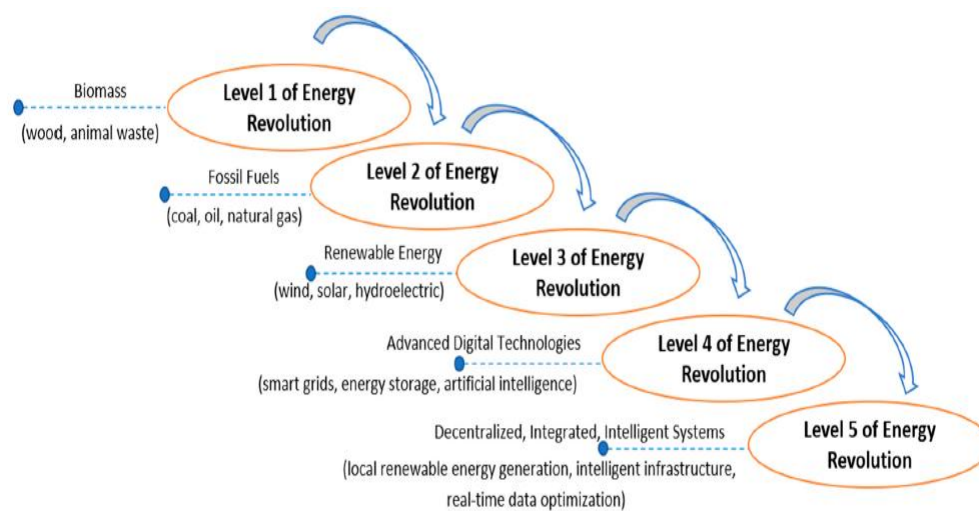


Figure 3. Evolution of the energy revolution (Levels 1 to 5).

